

Task 17 — Customer Segmentation (RFM)

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Mastery · Expert

Level 17 of 20 · ramp 81%

1 · What this is & why it matters

Segment customers using RFM (Recency, Frequency, Monetary) to target and prioritise them by value.

Not all users are equal. RFM turns one undifferentiated base into actionable segments for retention and growth.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: An RFM segmentation with named segments, sizes, value and recommended actions.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Recency

how recently they acted; Frequency -- how often; Monetary -- how much value.

Scoring & binning

ranking users into tiers on each dimension.

Segment naming

turning scores into meaningful groups (champions, at-risk, lost).

Actionability

each segment paired with a treatment.

Segment stability

whether segments hold over time.

5 · Recommended stack & tools

- pandas (qcut/rank)
- seaborn
- a segment summary table

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Compute recency, frequency and monetary values per customer.

2. Score each dimension into tiers (e.g. quintiles).
3. Combine scores into named segments.
4. Profile each segment's size and value.
5. Recommend an action per segment.
6. Validate that segments are distinct and stable.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Rule-based RFM vs clustering-based segmentation.
- Static segments vs rolling re-scoring.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- Which segment is most undervalued today?
- What single action would move 'at-risk' back to 'active'?
- Do the segments actually differ in behaviour, or only on paper?

9 · Definition of Done

Done when all are true and demoable

- An RFM segmentation with named segments, sizes, value and recommended actions.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — An RFM segmentation with named segments, sizes, value and recommended actions.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Segments with no attached action.
- Equal-width bins on skewed spend.
- Never refreshing, so segments rot.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation

- Storytelling with data and dashboard design
- Time-series and segmentation techniques